External Validity of Risk Elicitation

Episode IV: Measurement Error

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Slovic (1962):

TABLE 1

INTERCORRELATIONS AMONG RISK TAKING MEASURES
($N = 82$)

Variable	1	2	3	4	5	6	7	8
Response sets								
1 Dot Estimation								
2 Word Meanings	−.17							
3 Test Risk	.16	.05						
Questionnaires								
4 Life Experience Inventory	.05	.27**	−.04					
5 Job Preference Inventory[a]	.07	−.14	−.19	−.06				
Gambling preferences								
6 Self-Crediting Test	−.08	.19*	−.24*	.05	.09			
7 Variance preferences	.32**	.03	−.07	.23*	.07	.04		
8 Probability preferences	.16	−.03	−.07	−.03	−.35*	−.20	−.17	
Ratings								
9 Risk rating	.05	.00	−.24*	.34**	.10	−.02	.02	.18*

> *"…future research must carefully consider the problem of adequately **defining** and **assessing** risk taking behavior."*

So, how are we doing?

we live 60 years in Slovic’s future

This talk

- **Setting the stage**
  - the state of the art in the 2020s
- **Moving forward**
  - measurement error

Setting the stage

What is *risk*?

risk noun

\ ŕisk \

Definition of *risk* (Entry 1 of 2)

1 : possibility of loss or injury : PERIL

2 : someone or something that creates or suggests a hazard

3 **a** : the chance of loss or the perils to the subject matter of an insurance contract

also : the degree of probability of such loss

b : a person or thing that is a specified hazard to an insurer

c : an insurance hazard from a specified cause or source

// war risk

4 : the chance that an investment (such as a stock or commodity) will lose value

Measuring risk attitudes

> A **difficult** task with **crucial** relevance

- directly *unobservable*: *latent* (⇒ requires a theory)
- should we..
  - *infer* from real world data or from *ad-hoc* choices
  - ask or **t**ask?
  - elicit by *description* or by *experience*?

The state of the art: psychology

> Risk as **probability of harm**.

- **Questionnaires**:
  - directly ask, over different domains
  - tackle risk perception
- **Tasks**
  - putting the subject in a ‘risky’ situation
  - card/gambling tasks

The state of the art: economics

> risk formally defined as **uncertainty over outcomes**.

The lottery paradigm

- incentives – choice over lotteries
- strong theoretical underpinning
- different formats, cover stories, contexts
- estimation of utility functions (⇒ models)

> Metric of success: **internal validity** (task ⇒ theory)

Tools: RETs

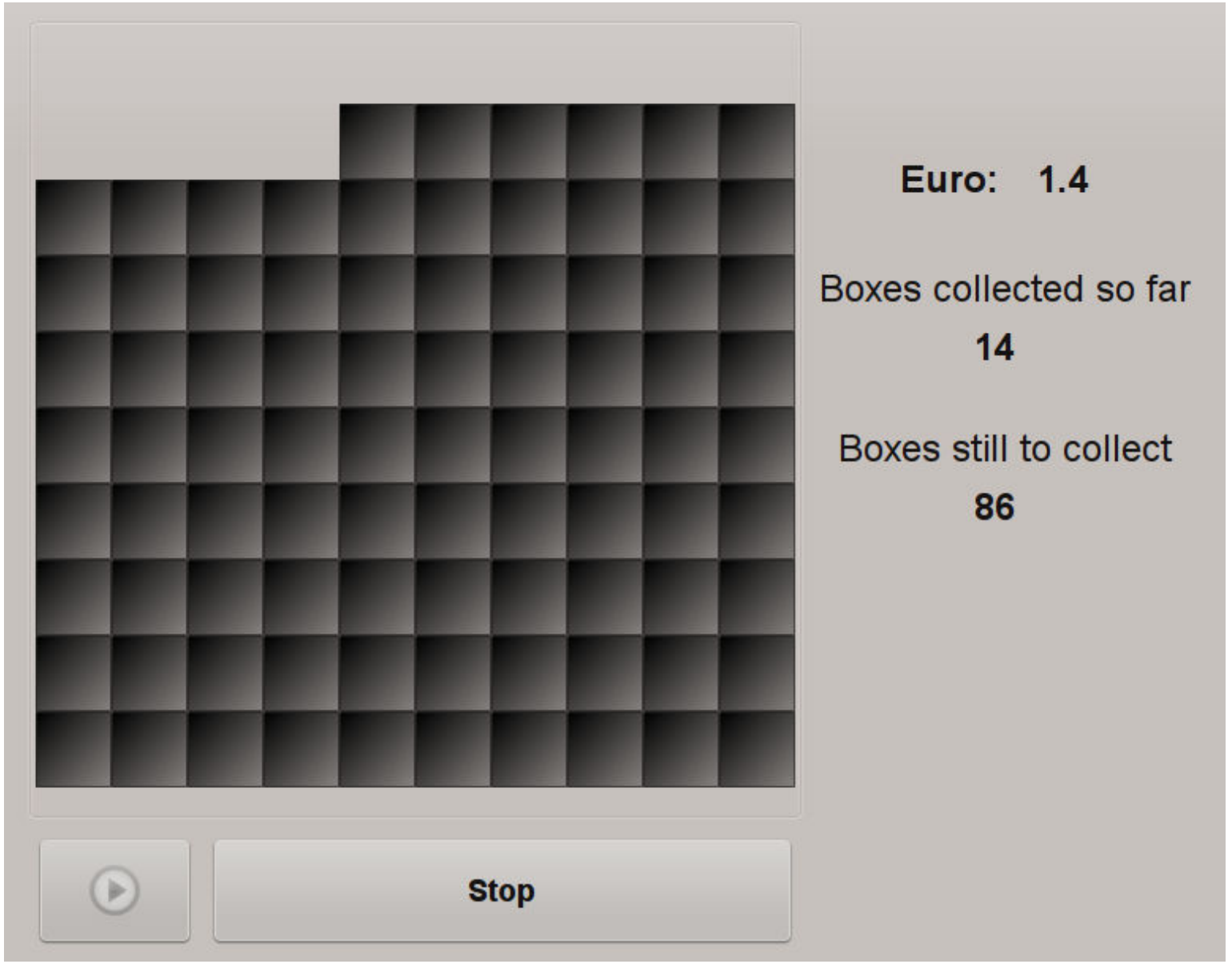
Holt and Laury

	Option A				Option B			
1	1/10	4 €	9/10	3.2 €	1/10	7.7 €	9/10	0.2 €
2	2/10	4 €	8/10	3.2 €	2/10	7.7 €	8/10	0.2 €
3	3/10	4 €	7/10	3.2 €	3/10	7.7 €	7/10	0.2 €
4	4/10	4 €	6/10	3.2 €	4/10	7.7 €	6/10	0.2 €
5	5/10	4 €	5/10	3.2 €	5/10	7.7 €	5/10	0.2 €
6	6/10	4 €	4/10	3.2 €	6/10	7.7 €	4/10	0.2 €
7	7/10	4 €	3/10	3.2 €	7/10	7.7 €	3/10	0.2 €
8	8/10	4 €	2/10	3.2 €	8/10	7.7 €	2/10	0.2 €
9	9/10	4 €	1/10	3.2 €	9/10	7.7 €	1/10	0.2 €
10	10/10	4 €	0/10	3.2 €	10/10	7.7 €	0/10	0.2 €

Binswanger / Eckel and Grossmann

	Event	Probability	Outcome
1	A	50%	4 €
	B	50%	4 €
2	A	50%	6 €
	B	50%	3 €
3	A	50%	8 €
	B	50%	2 €
4	A	50%	10 €
	B	50%	1 €
5	A	50%	12 €
	B	50%	0 €

Bomb Risk Elicitation Task



Investment Game (Gneezy & Potters)

Endowment X

How much would you like to invest?

Safe account
1 : 1

Risky investment
1 : {1/2: 2.5; 1/2: 0}

Certainty Equivalent MPL

A	B	
100%	50%	50%
0		
10		
20		
30		
40		
50	100	0
60		
70		
80		
90		
100		

Questionnaire: SOEP

How likely are you to take risks in general, one a scale from 0 (not taking any risks) to 10 (taking many risks)?

(with further additional questions by domain, as health, driving, sports…)

Questionnaire: DOSPERT

Domain **Spe**cific **R**isk **T**aking Scale

- 6 domains: investing, gambling, health/safety, recreational, ethical, and social
- 1 to 7 scale: *how risky do you think X is?*
- 1 to 7 scale: *how likely are you to engage in X?*

Examples:

- Riding a motorcycle without a helmet.
- Investing 10% of your annual income in a moderate growth diversified fund.

What do we know?

METARET

> A meta-analysis of Risk elicitation tasks

- **elicited** risk attitudes: tasks and questionnaires
- **convergent** validity: correlations among measures
- **predictive** validity: correlation task $\iff$ questionnaires

Explore the data!

> Live data exploration on a shiny app

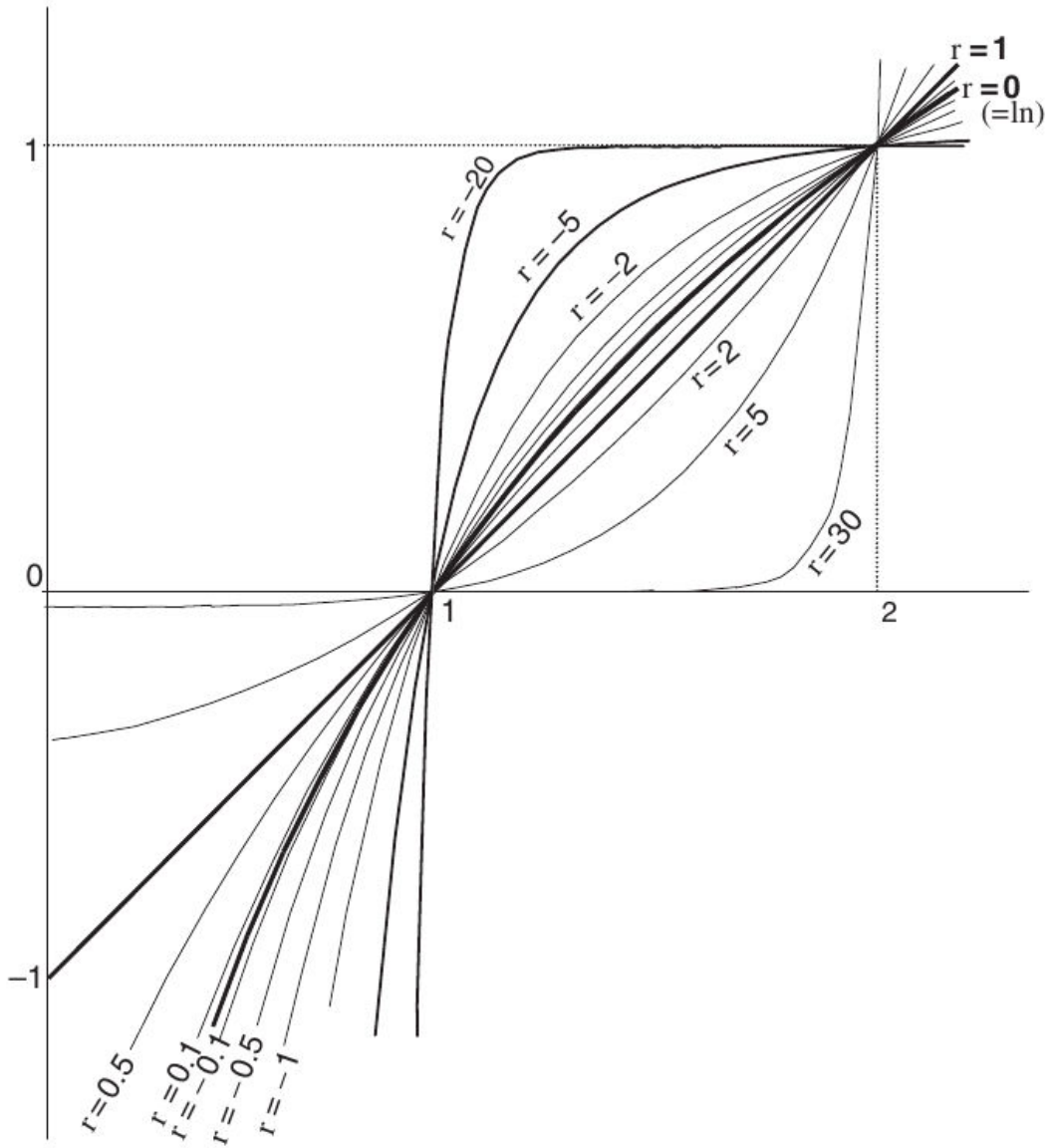


Assumptions: CRRA (à la Wakker)

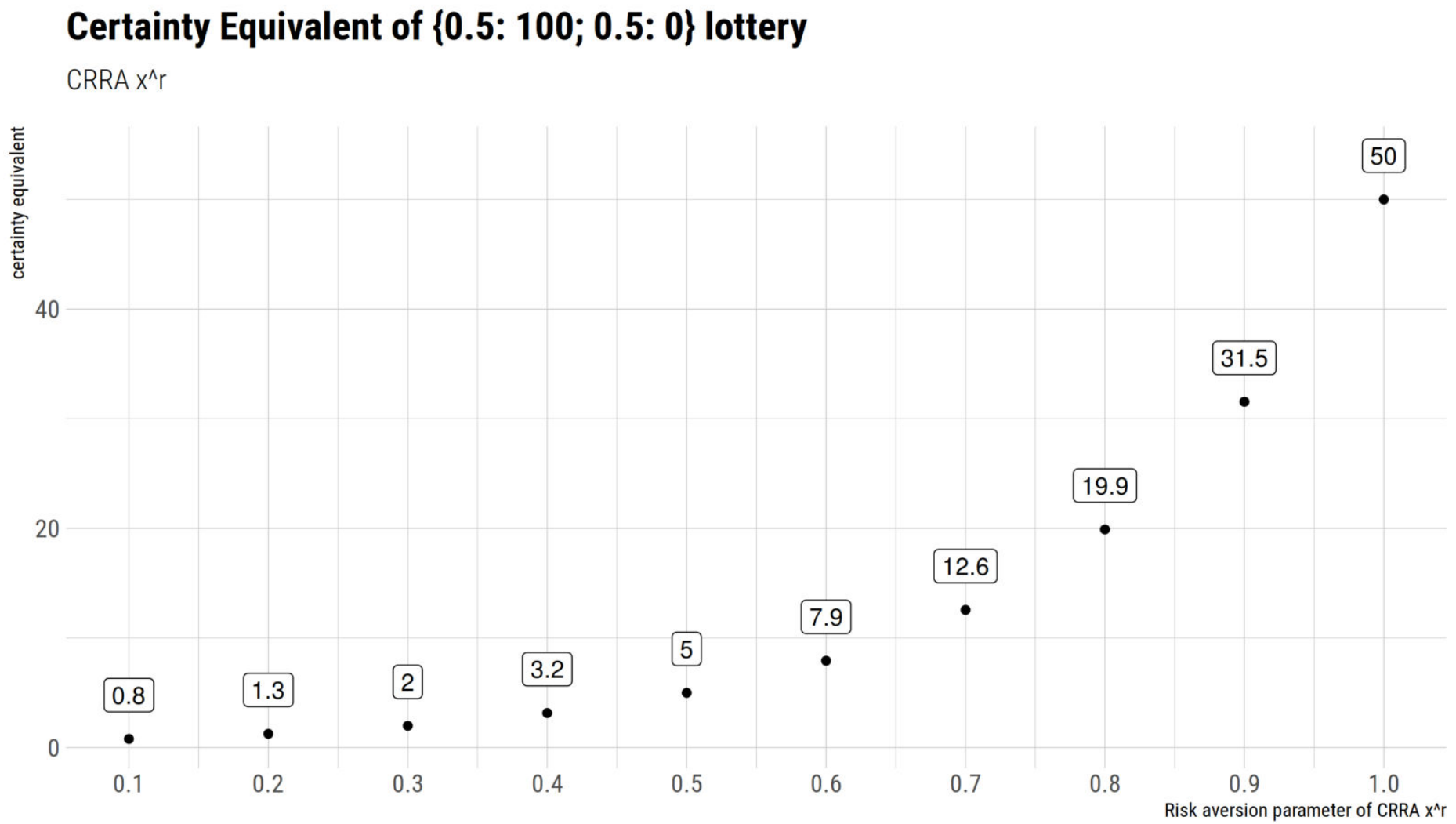
$$u(x) = x^r$$

- simple
- captures risk aversion
- makes different tasks comparable

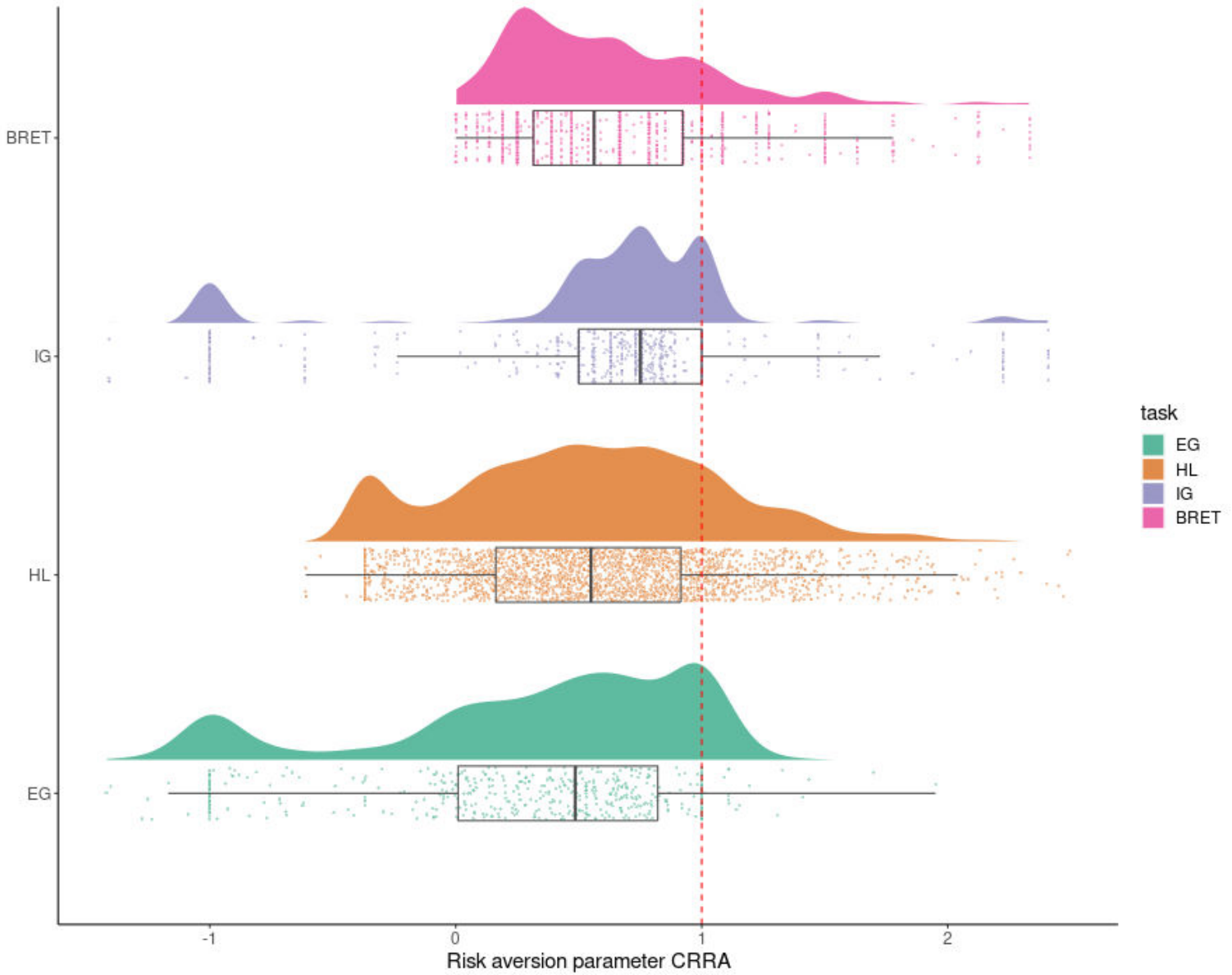
CRRA



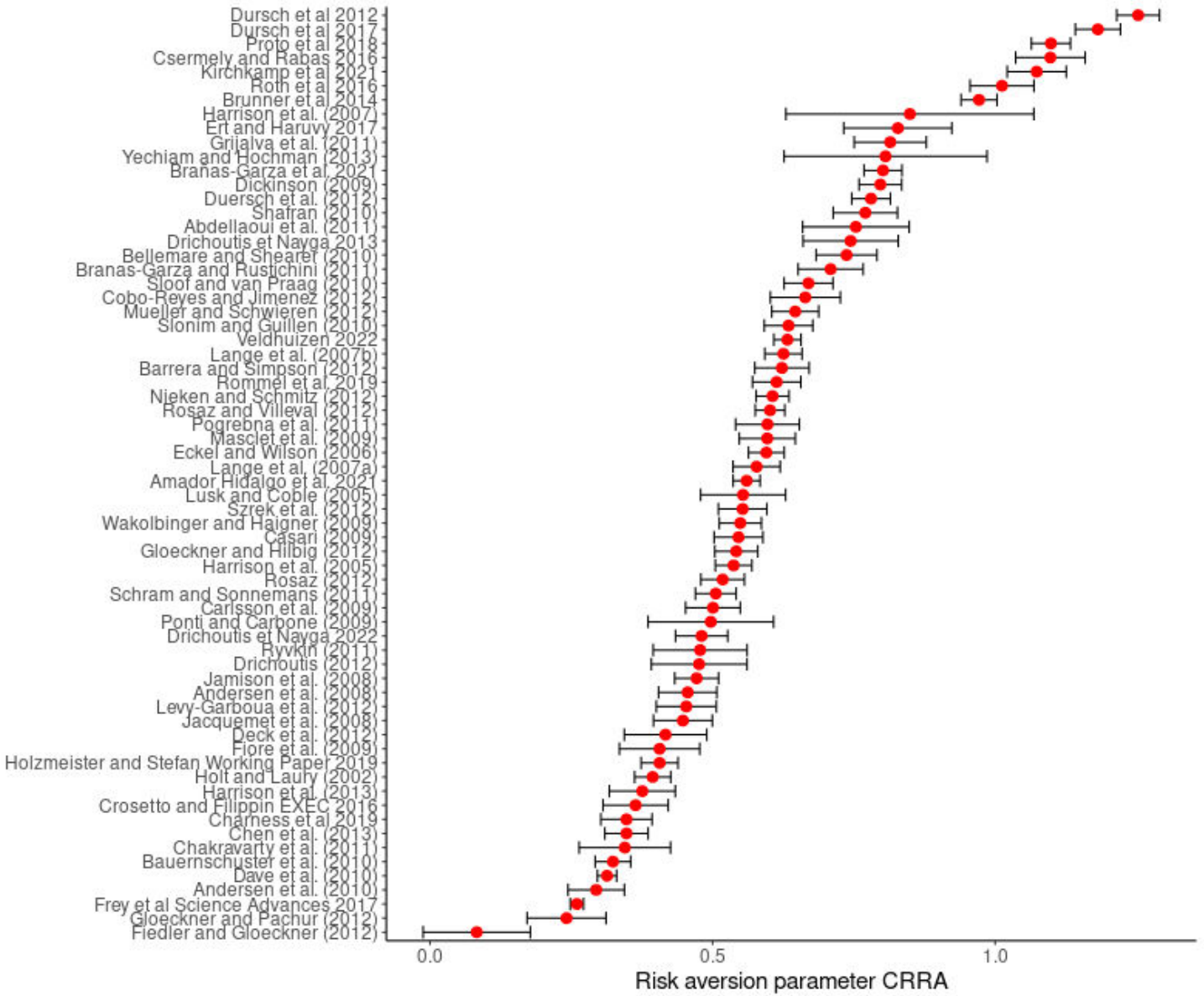
How big are the differences?



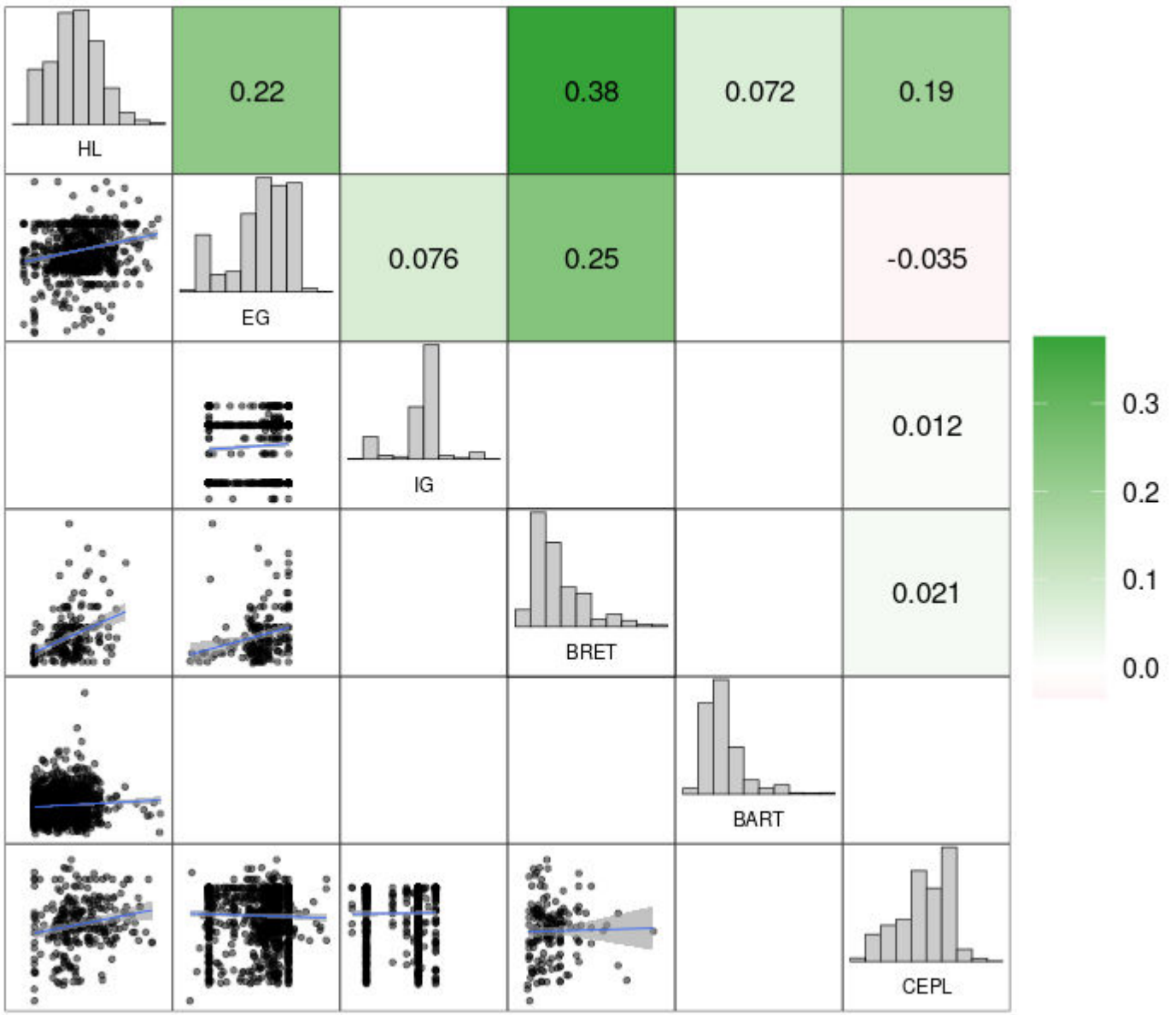
Low consistency *across* tasks



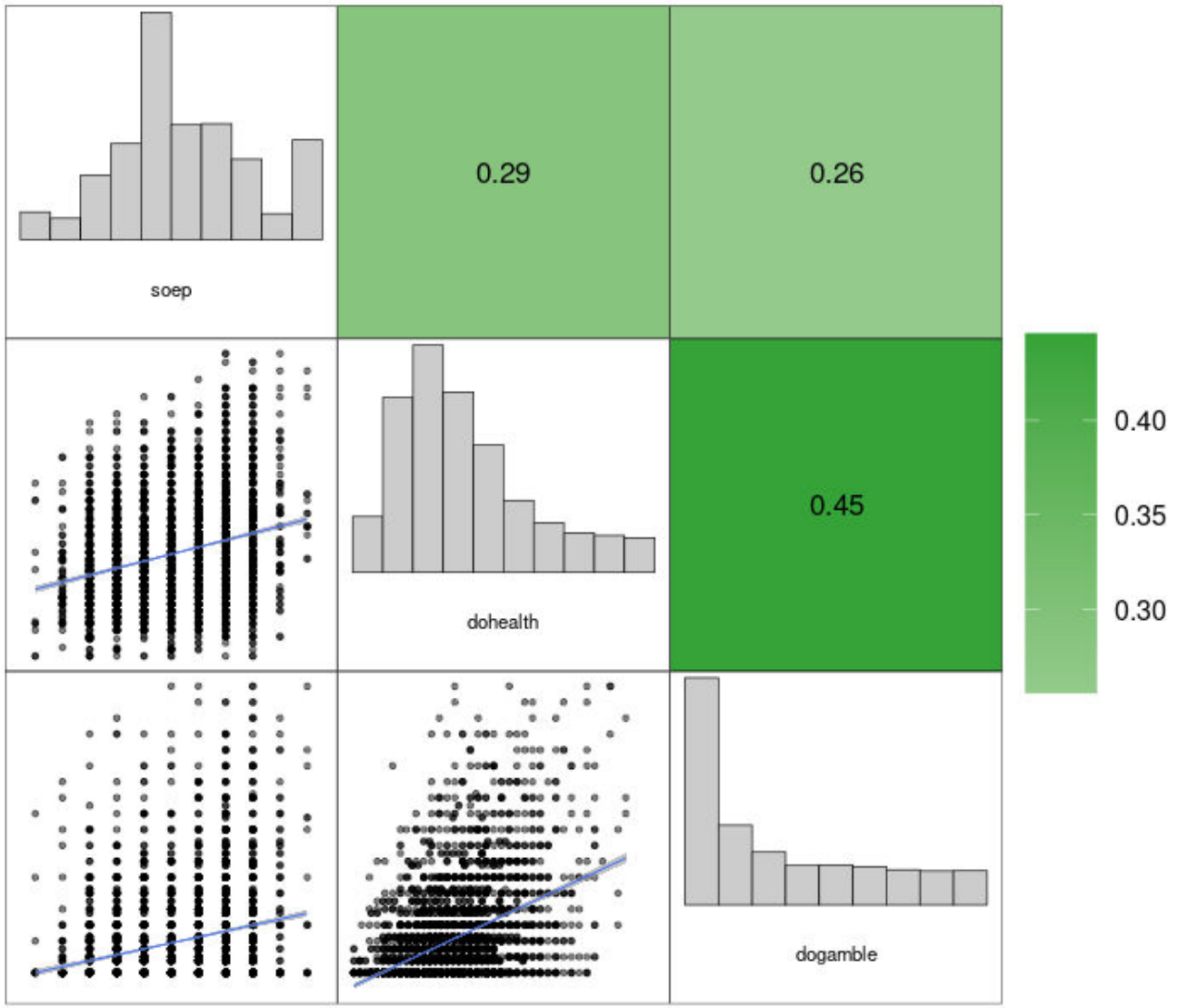
Low consistency *within* tasks



Convergence: tasks



Convergence: questionnaires



Tasks ⟺ Questionnaires



Summing up…

- *"…future research must carefully consider the problem of adequately **defining** and **assessing** risk taking behavior."*

- **exactly as in 1962**


I am your father.

The road ahead

Is it *perception*?

> *observed* = *attitude* + *perception* ⇒ biased inference

risk noun

\ ˈrisk \

Definition of *risk* (Entry 1 of 2)

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4 : the chance that an investment (such as a stock or commodity) will lose value

Is it *noise*?

> *noisy* preference + one-shot choices ⇒ noisy data

- fuzzy preferences
  - i.i.d. ⇒ measurement error
  - *not* i.i.d. ⇒ task-specific bias (*Crosetto & Filippin 2015*)
- other potential reasons for noise
  - cognitive limits ⇒ limited understanding
  - context-dependence

Measurement error

(with Antonio FIlippin, Daniel Navarro Martinez, Xinghua Wang)

Some references to get there

> Measurement error: $\hat{X} = X + \varepsilon, \ \varepsilon \text{ i.i.d.}$

- *(1) Gillen et al 2019* – ME $\Rightarrow$ false positives + techniques
- *(2) Galizzi & Navarro Martinez 2019* – social preference games have low external validity
- *(3) Navarro Martinez & Wang 2022* – applying (1) to (2) increases external validity
- *(4) this paper* – applying (3) to the risk elicitation puzzle

Goal

> Does aggregating **multiple** measures **over time** to reduce measurement error help solving the *risk elicitation puzzle*?

Experimental design

Setup

> Follow subjects for **2 weeks** taking **repeated** measures of

- Questionnaires;
- Risk Elicitation Tasks;
- Daily Reconstruction Method (*Kahneman et al. 2004*)

Timeline

- **Day 0**: lab session, instructions, all tasks, all questionnaires
- **Days 1-14**
  - every day: Daily Reconstruction Method
  - every odd day: Tasks
  - every even day: Questionnaires

> **Payment**: one random tasks in day 0 + show-up fee; 1 task per odd day + 1.5€ per DRM

Validity checks

> We test different type of psychometric validity

- *Test-retest reliability* for tasks and questionnaires
- *Convergent validity* among tasks
- *Convergent validity* among questionnaires
- *Convergent validity* tasks <> questionnaires
- *External validity* task > DRM and questionnaires > DRM

Questionnaires

> Focus on the most widely used

- **SOEP** (aggregate and by domain, 1 question)
- **DOSPERT** (over domains, 30 questions)

Tasks

> Easy, intuitive, fine-grained + **loss aversion**

- **BRET** (intuitive, fine-grained, “game”)
- **HL** (more complex, “standard”)
- **Investment game** (intutive, fine-grained, investment)
- **Loss** (we need a measure of loss aversion)

Loss task

> Estimating λ

- estimate r from first three tasks
- assume $r^+ = r^-$
- provide a price list that identifies λ

Loss task

Decision	Option C			Option D			
	Gain			Prob	Gain	Prob	Gain
1	0,0€	○	○	*50%*	10,0€	*50%*	-13,3€
2	0,0€	○	○	*50%*	10,0€	*50%*	-10,0€
3	0,0€	○	○	*50%*	10,0€	*50%*	-8,0€
4	0,0€	○	○	*50%*	10,0€	*50%*	-6,7€
5	0,0€	○	○	*50%*	10,0€	*50%*	-5,7€
6	0,0€	○	○	*50%*	10,0€	*50%*	-5,0€
7	0,0€	○	○	*50%*	10,0€	*50%*	-4,4€

DRM

> List of all the *active* decisions under risk of the day

Filled from 6pm to midnight, every day:

- **Domains**: health, safety, recreation, driving, financial, ethical, social
- **Do vs Avoid**: *not* taking risk is a decision under risk!

DRM

> List of all the *active* decisions under risk of the day

For each activity:

- category
- importance of positive (1..10) and negative (1..-10) consequences
- likelihood of positive and negative consequences (0..100%)
- perception of the risk avoided or taken when deciding (-10..0..10)

Privacy and credibility concerns

> We need our subjects to tell us the **truth**. That's not easy.

- subjects know only *one* person will read their raw descriptions
- that person does not and will never know their identity
- data cleaned from any identifying element by that person
- then shipped to 4 **external judges** for rating
- judge ratings, not subject's description, will be used & released

External judges

> We hire 4 “judges” to rate the overall risk taking by subjects

- for each activity, they fill the same questions as subjects
  - is the activity risky?
  - category
  - risk avoided/taken (-10..0..10)
- judges are paid lump-sum for their work
- 4 PhD students in economics, across France

(pre-registered) Hypotheses

> Measurement error **plays** a role: as we aggregate *more*…

- Test-retest reliability **up**
- Between-tasks convergence **up**
- Between-questionnaires convergence **up**
- Task-questionnaire convergence **up**
- External validity (against DRM) **up**

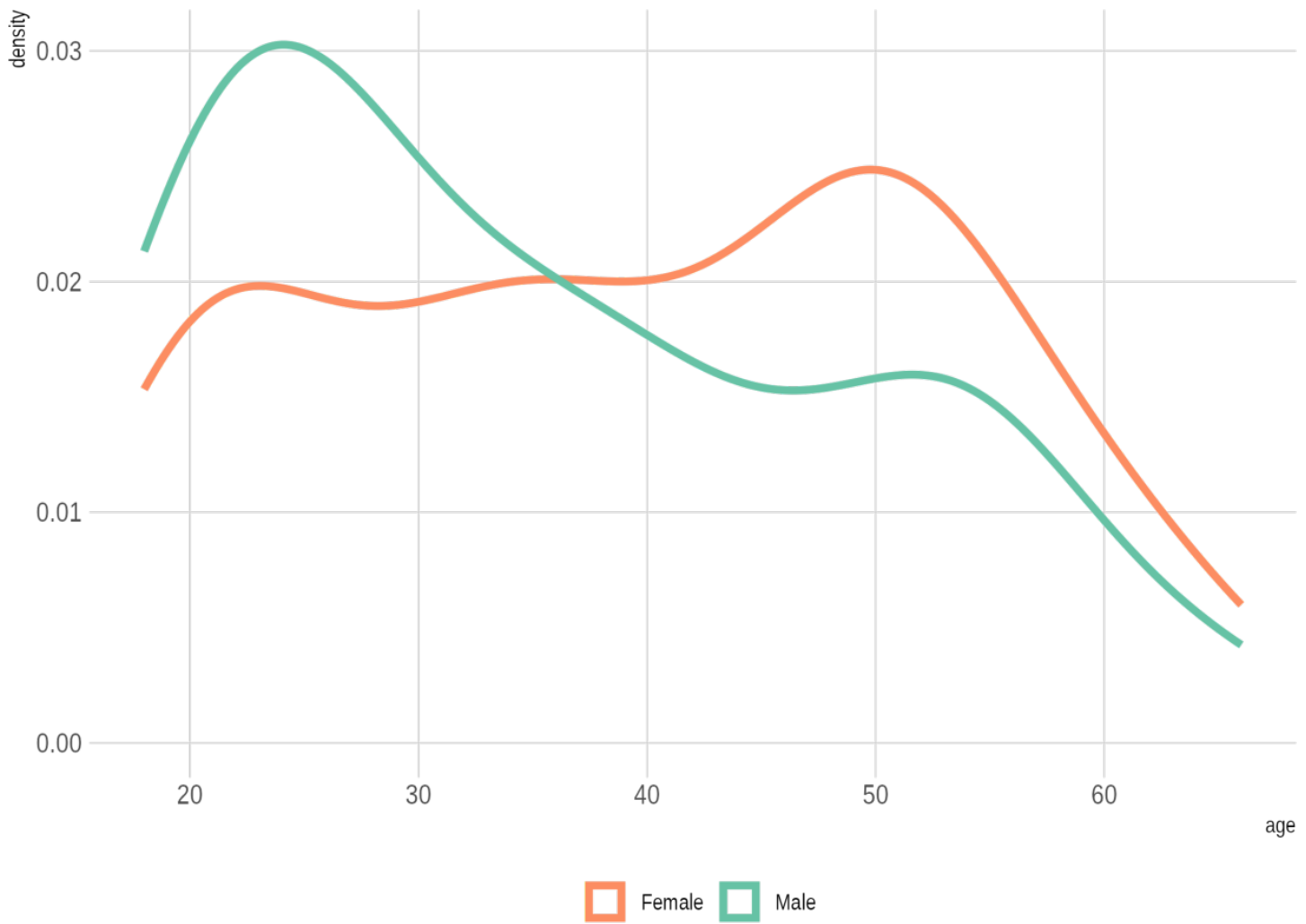
Results

Sample

- **161** subjects over 5 sessions
- **General** population
- Very **low** attrition over 14 days: **2 dropout** / **161**
- Average payment ~**77€** (for the 14 days)

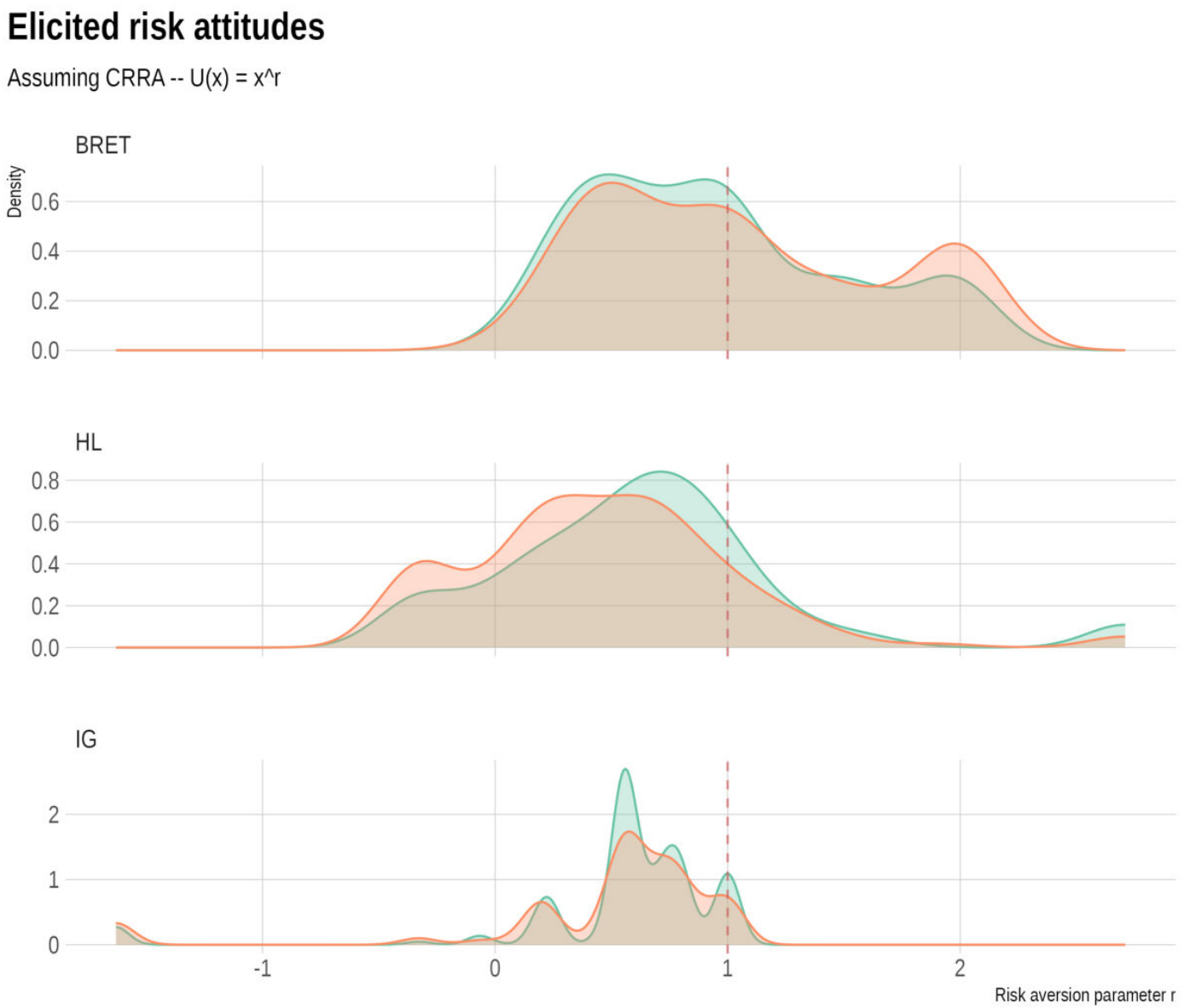
GENDER	N	AGE
f	100	39.8 (13.8)
m	59	35.4 (13.6)

Age distribution by gender

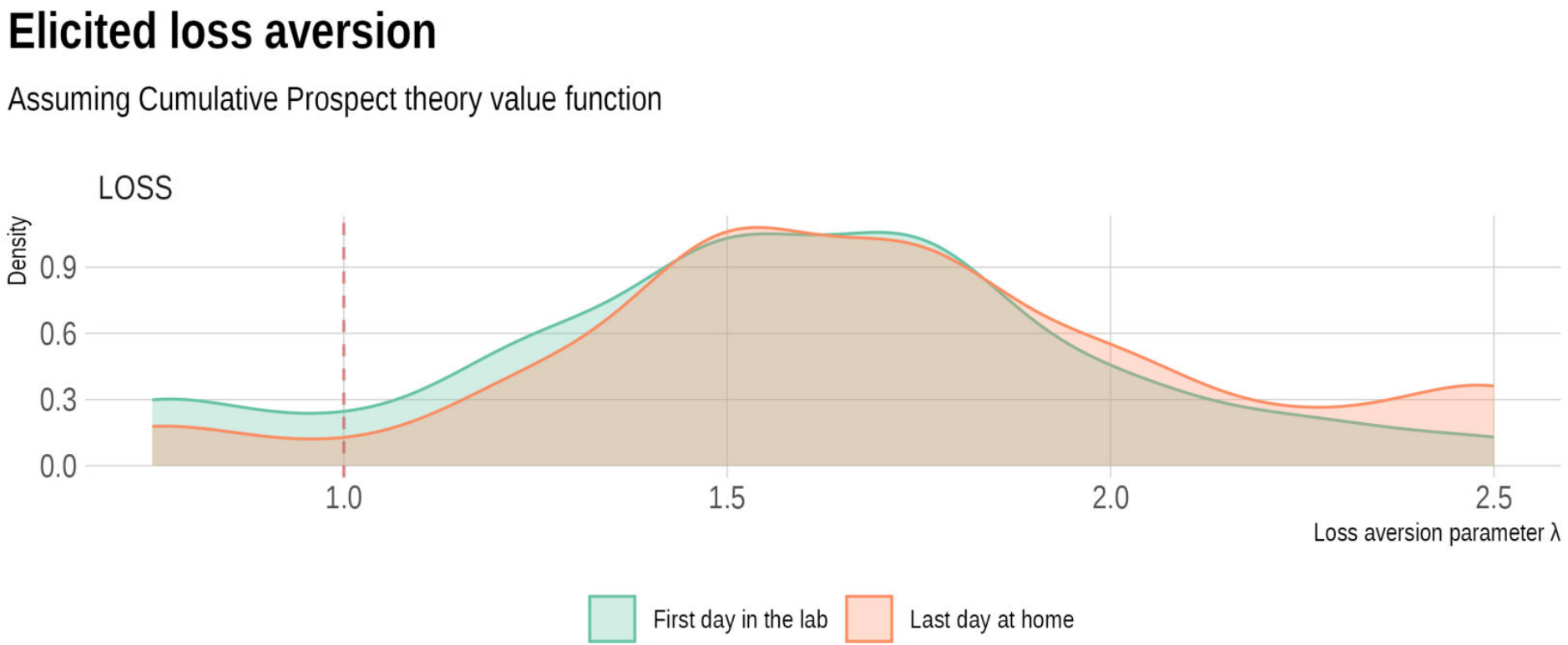


Test-retest

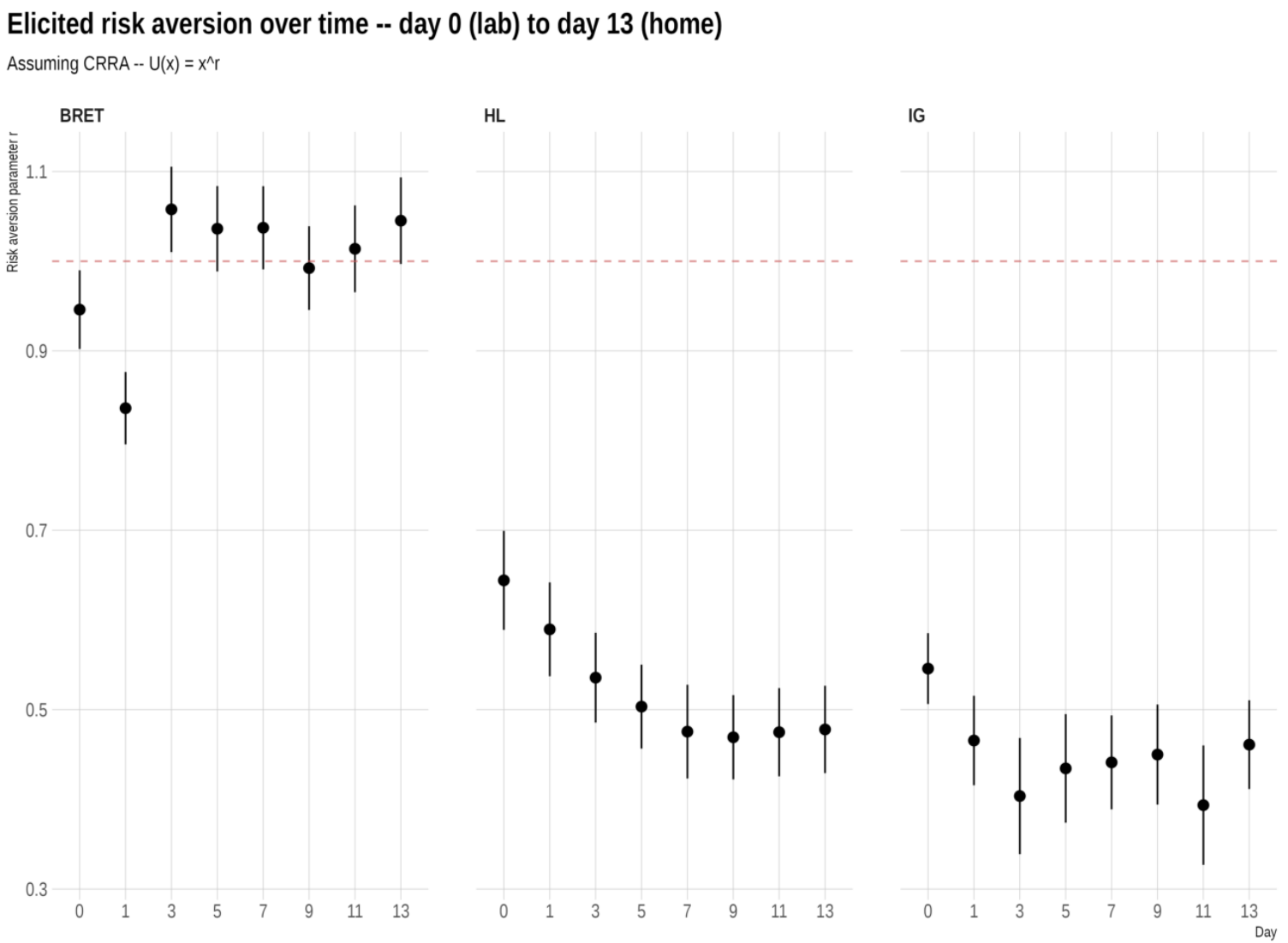
Elicited risk attitudes



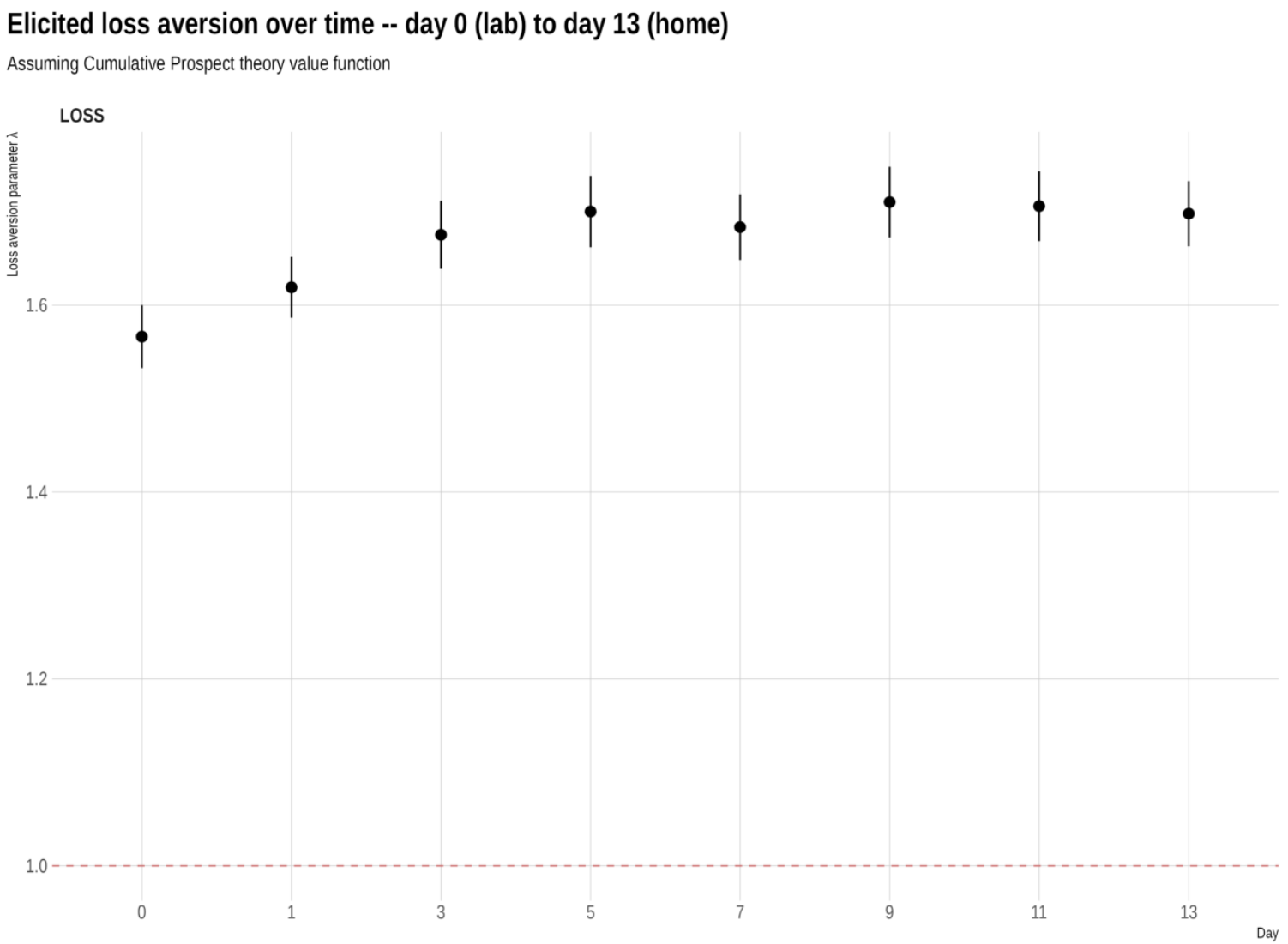
Elicited loss attitudes



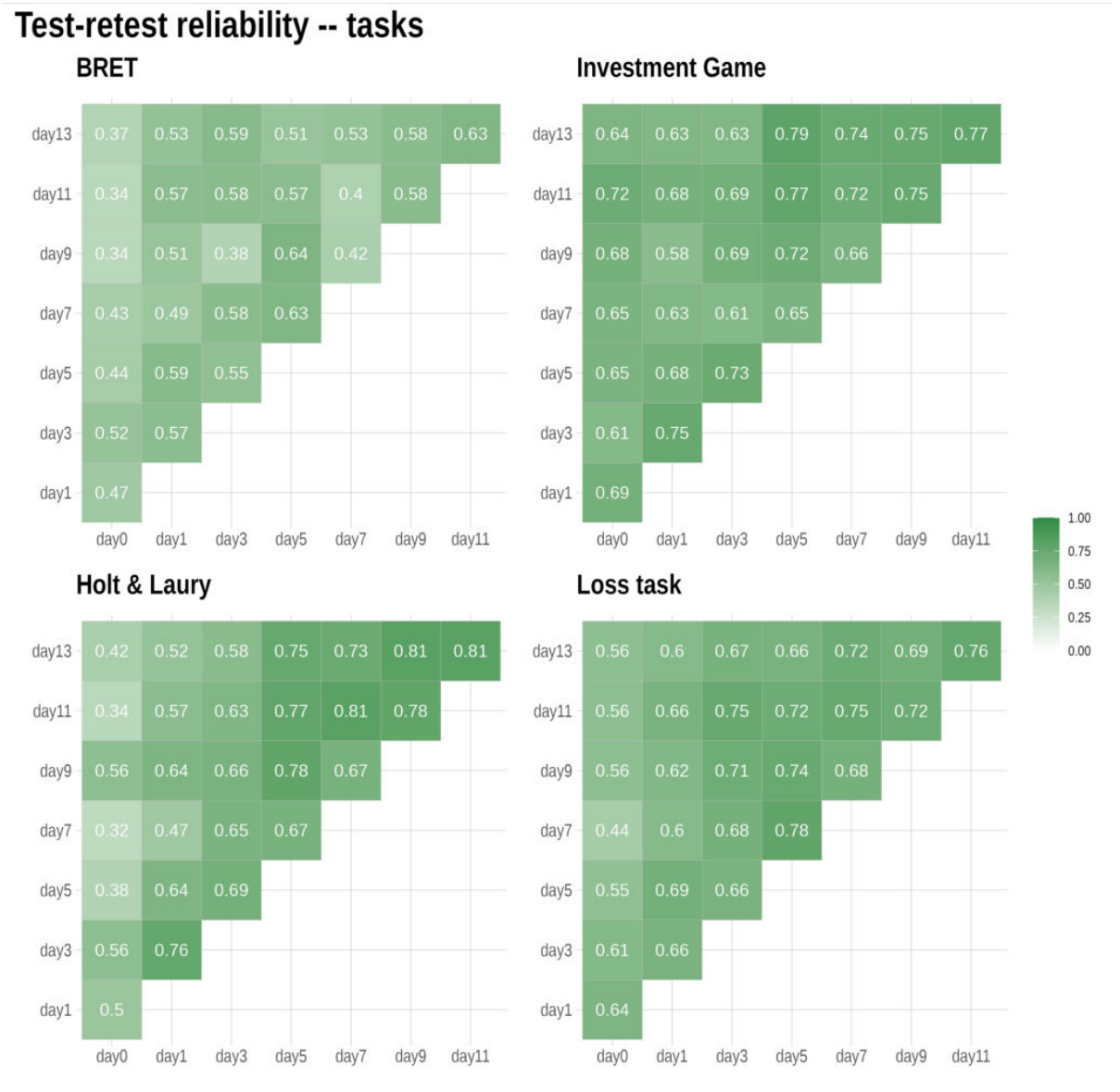
Evolution over 14 days: risk



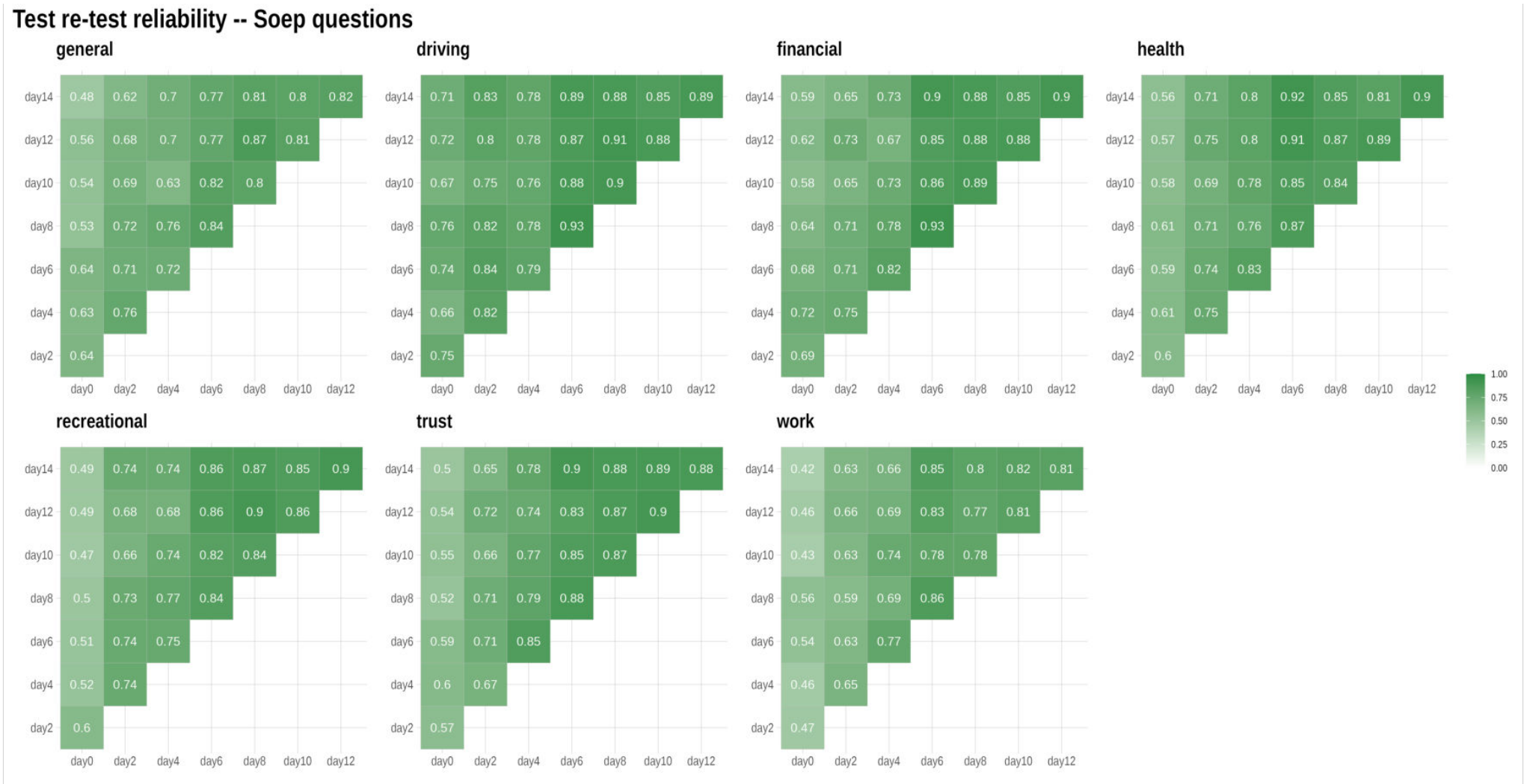
Evolution over 14 days: loss



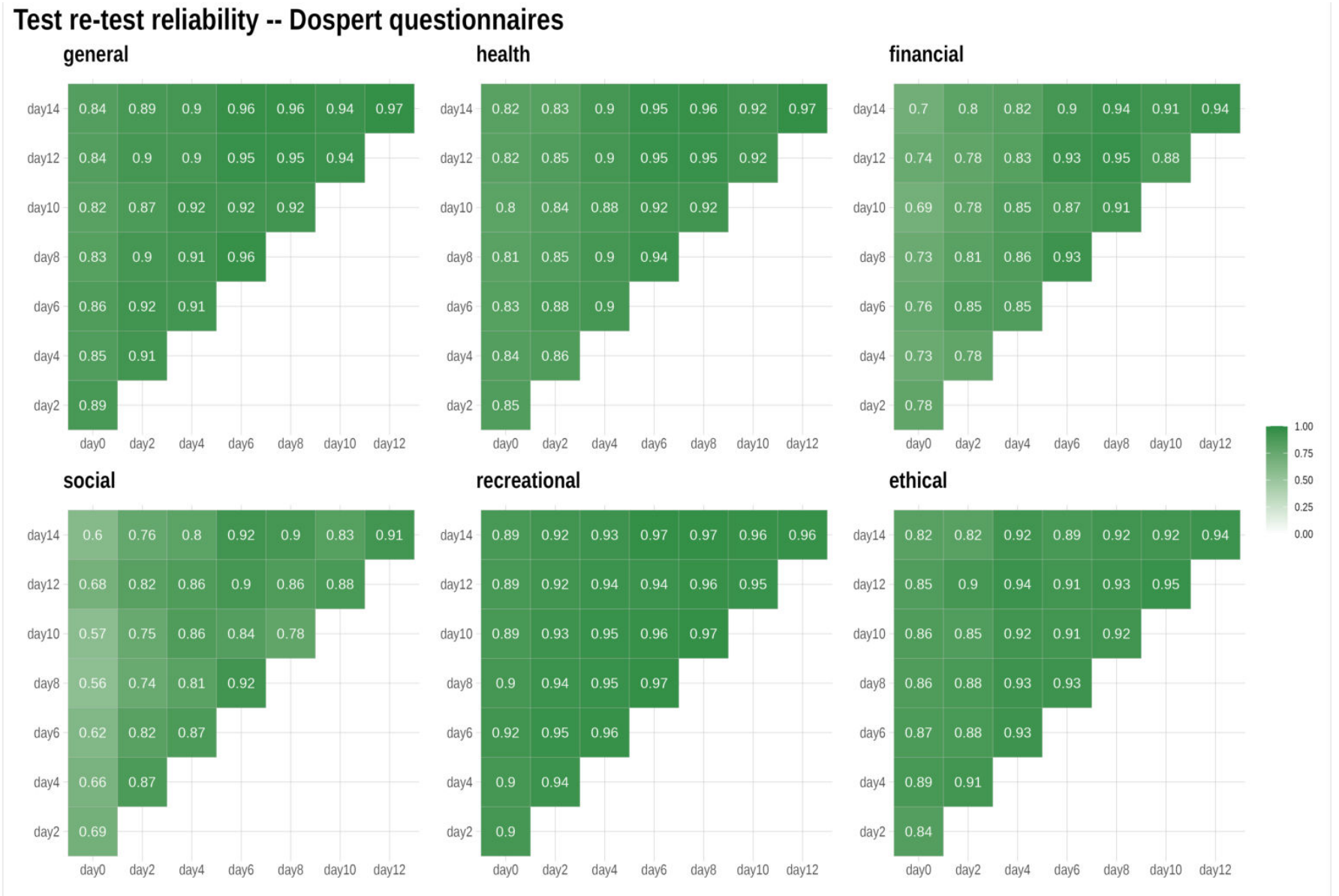
Test-retest reliability: tasks



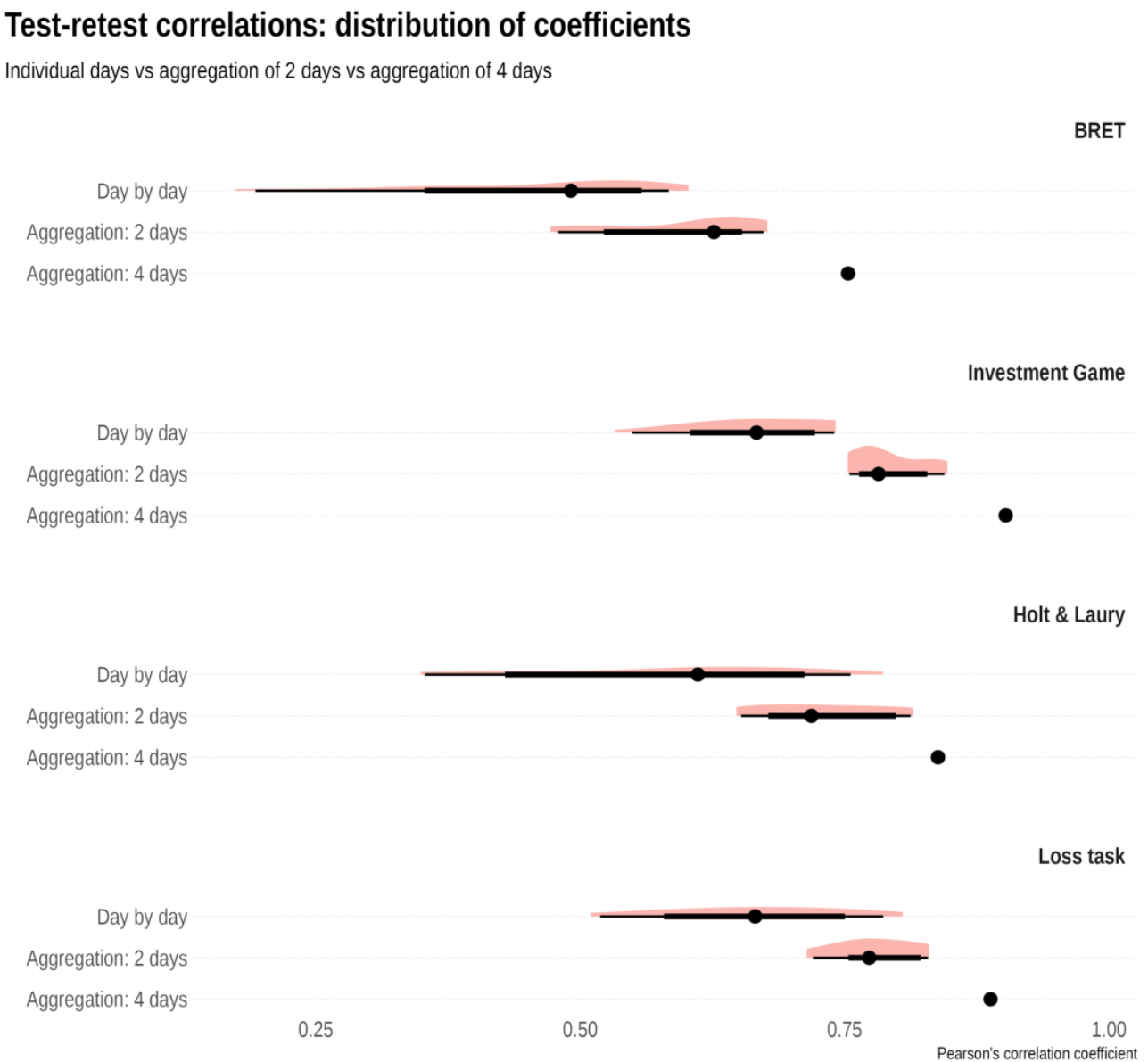
Test-retest reliability: SOEP



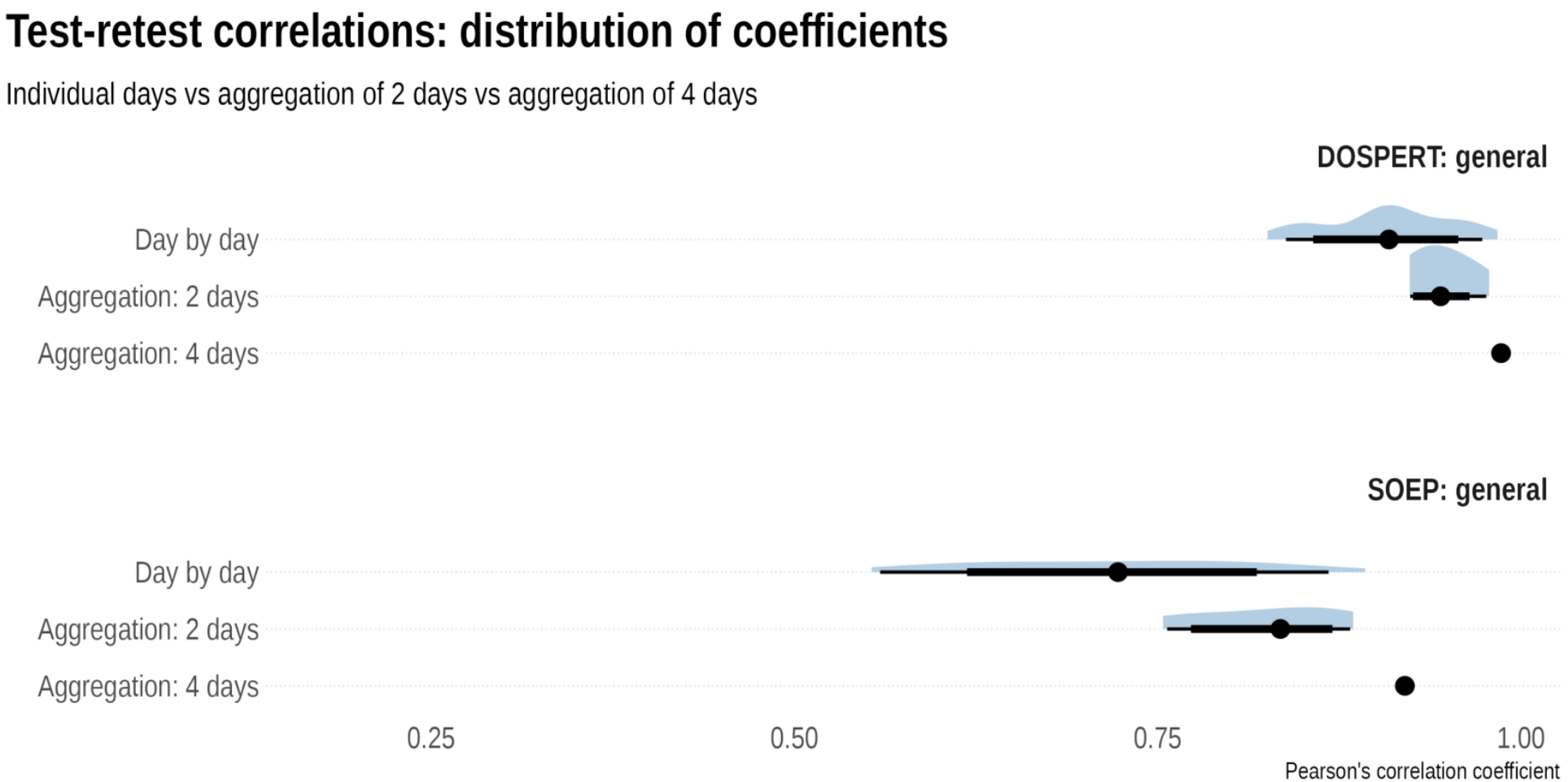
Test-retest reliability: DOSPERT



Test-retest: aggregating tasks

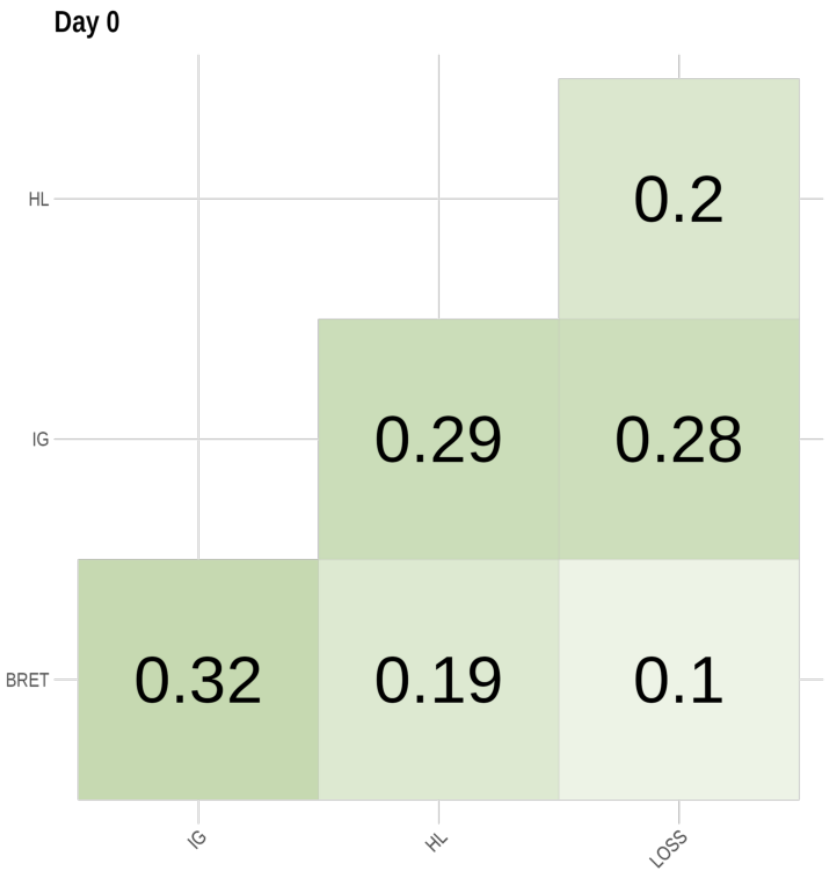


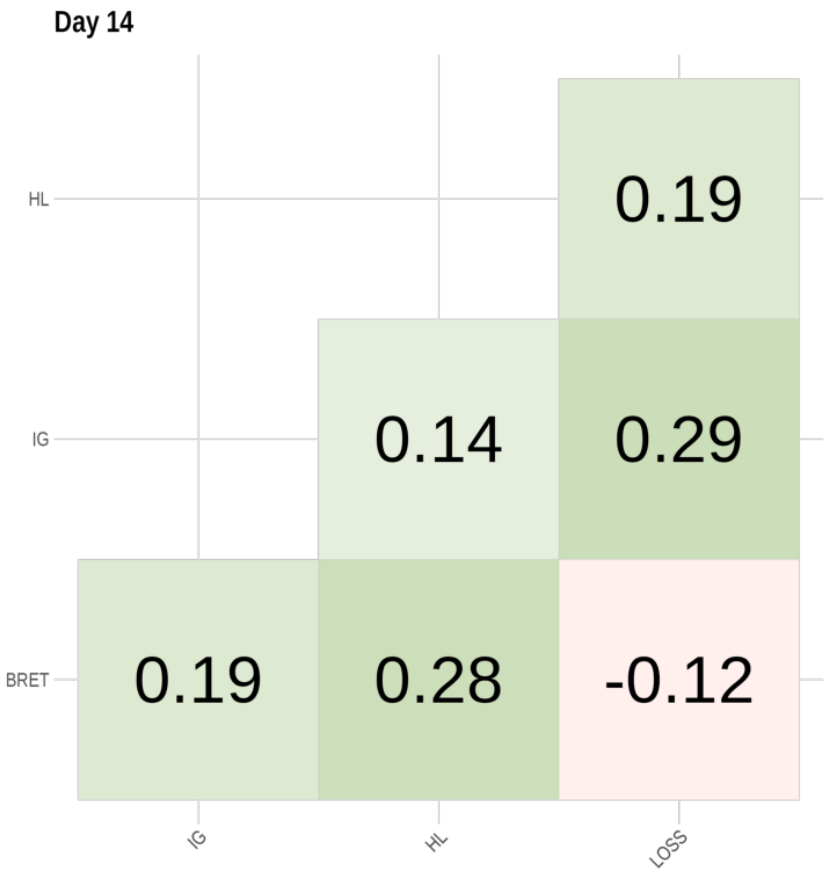
Test-retest: aggregating questionnaires



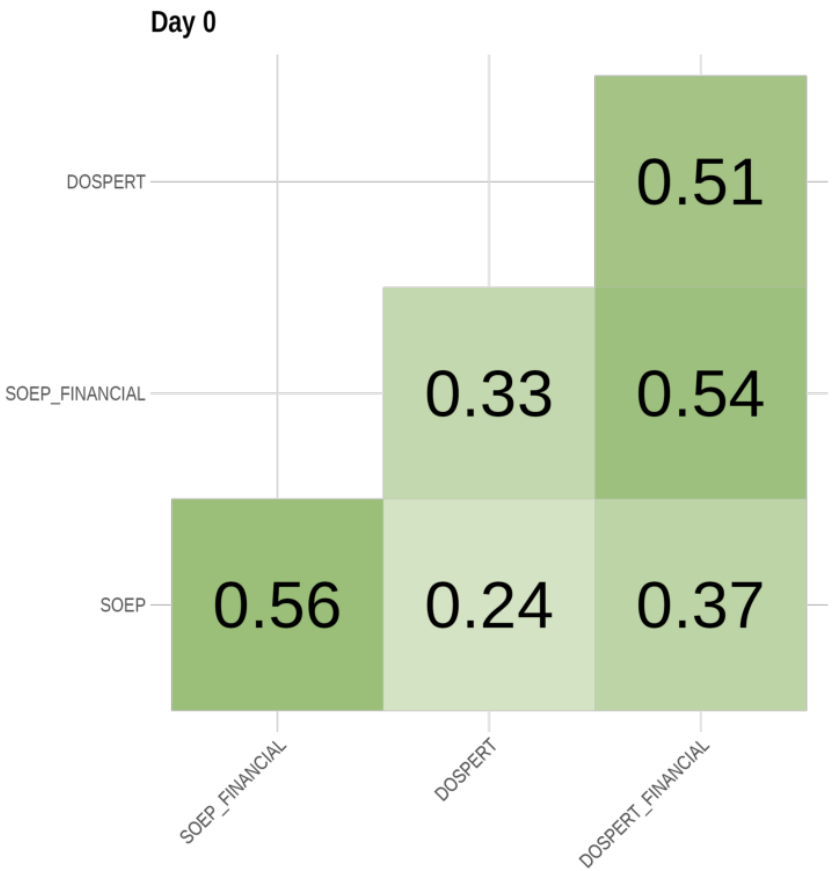
Convergent validity

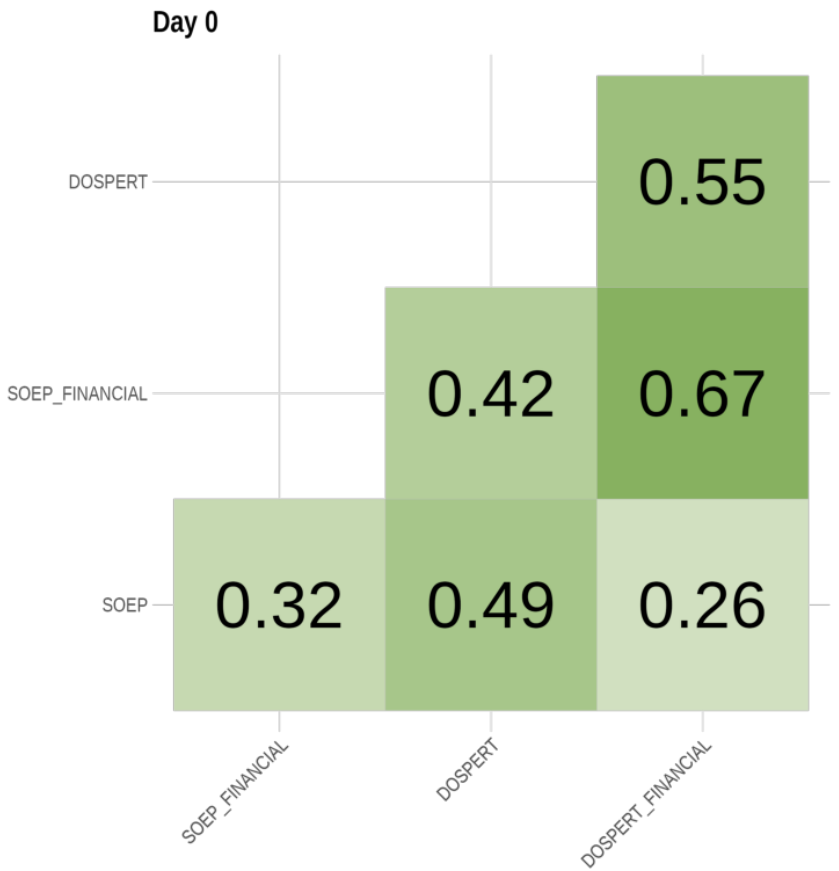
Convergence: tasks



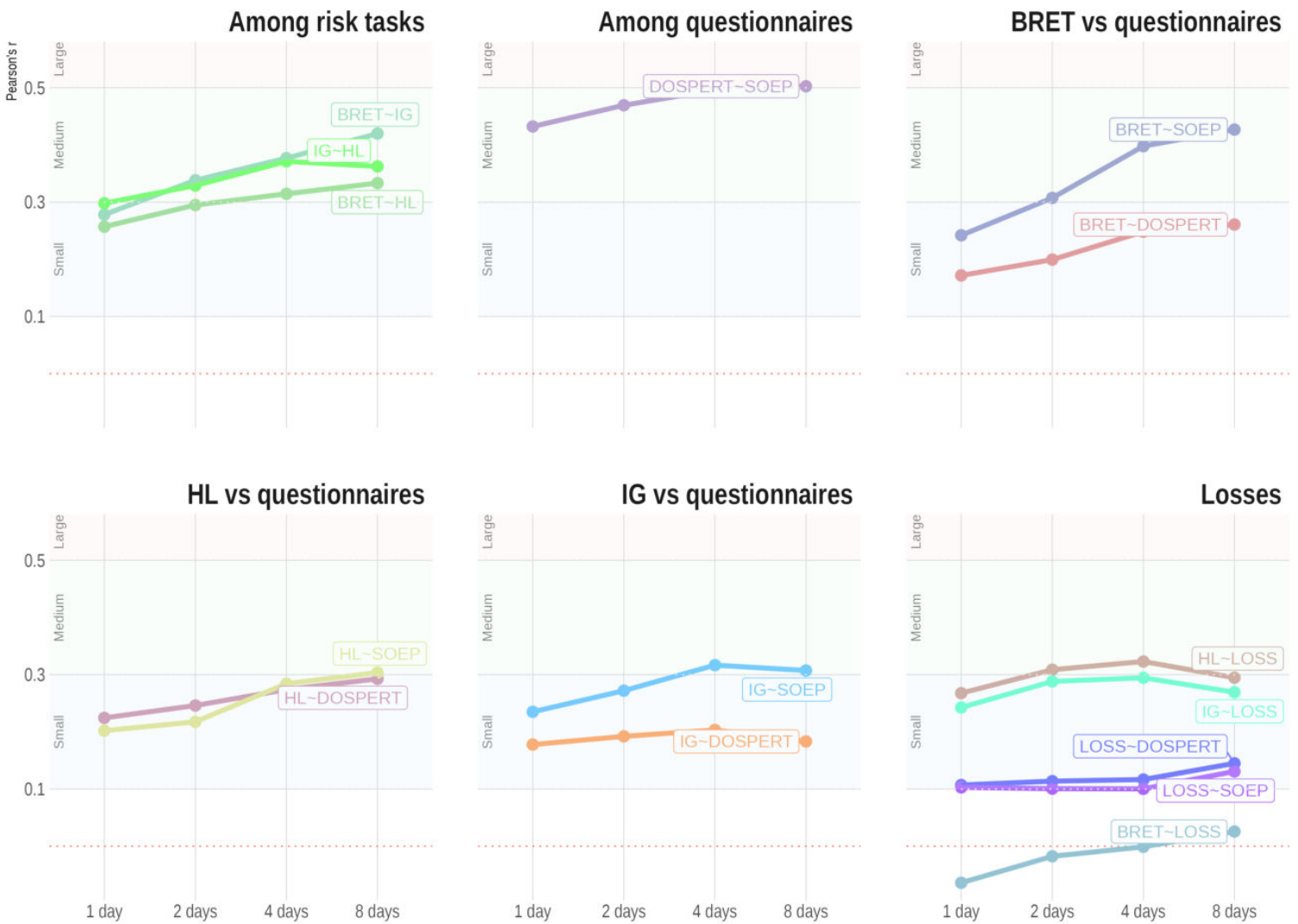


Convergence: questionnaires





Convergence: aggregating



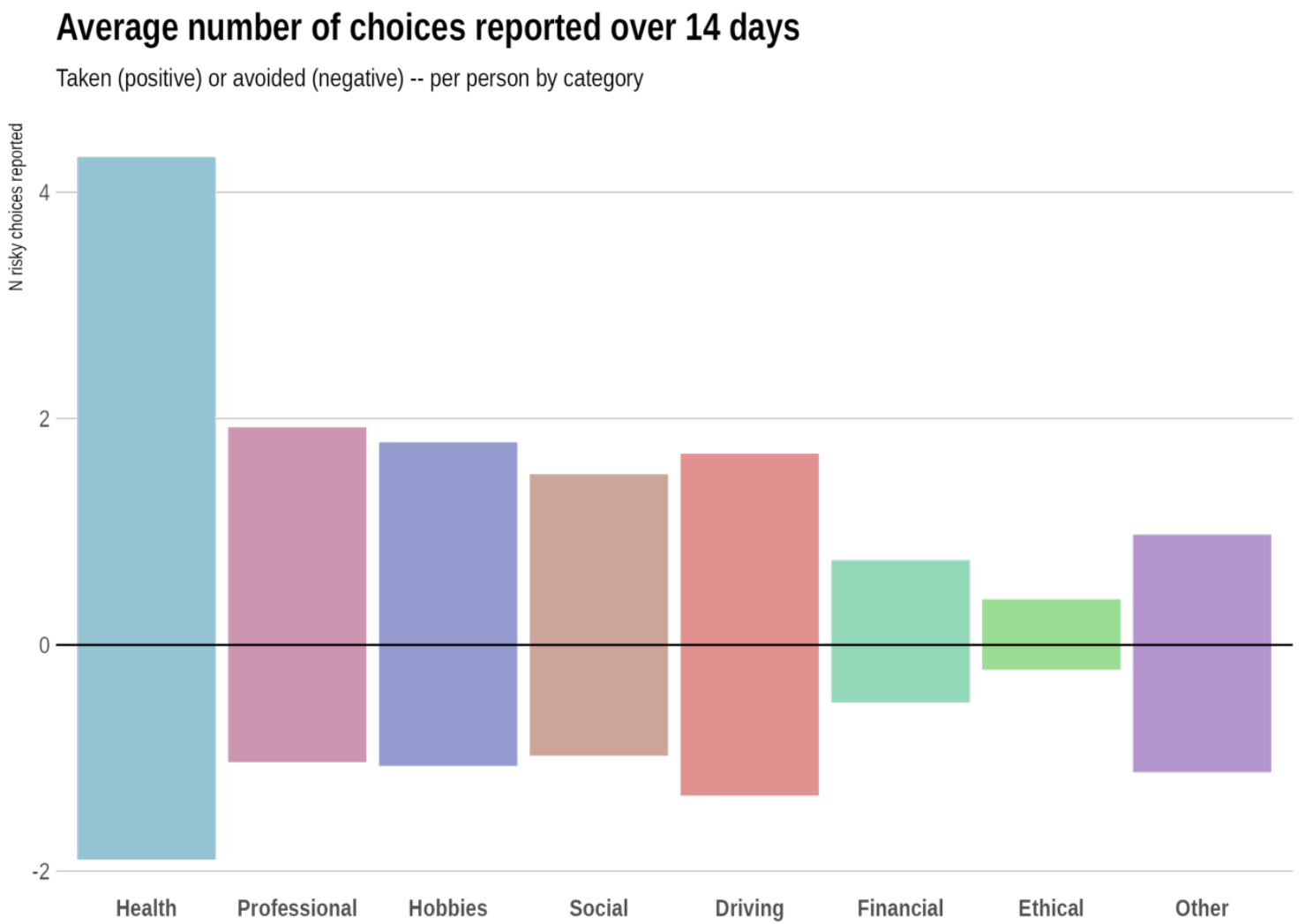
External validity: DRM

Descriptives

> 159 subjects carried out for 14 days, **3392** activities

- Subjects reported 1.52 choices involving risk per day
- 0.95 per day, they took risks – 0.57 per day, they avoided risks
- Mostly in **Health**, followed by **Driving, Professional** and **Hobbies**
- **Financial** and **Ethical** risk less frequent (<1/person over 2 weeks)

Descriptives



Measures

- **Absolute** frequency measure:

  $$\sum_{d=1}^{14} N_take_d$$

- **Relative** frequency measure:

  $$\sum_{d=1}^{14} \frac{N_take_d}{N_take_d + N_avoid_d}$$

- **Weighted** measure: average risk rating

  $$\frac{1}{14} \sum_{d=1}^{14} W_d,\ w \in [-10, 10]$$

Measures' validity



Judges have a higher bar for risk

> Subjects report **3392** choices in presence of risk…
>
> …but jdges think that many of them ***are not*** risky

- **Judge 1:** 2376/3992 (59%)
- **Judge 2:** 2411/3992 (60%)
- **Judge 3:** 2369/3992 (59%)
- **Judge 4:** 1992/3992 (49%)

Judges' reliability

Judge1	Judge2	% agree	Kappa	Agreement
Overall	–	55.7%	0.45	Moderate
Judge 1	Judge 2	77.7%	0.46	Moderate
Judge 1	Judge 3	75.6%	0.42	Moderate
Judge 1	Judge 4	69.3%	0.34	Fair
Judge 2	Judge 3	81.2%	0.55	Moderate
Judge 2	Judge 4	75.3%	0.47	Moderate
Judge 3	Judge 4	75.8%	0.48	Moderate

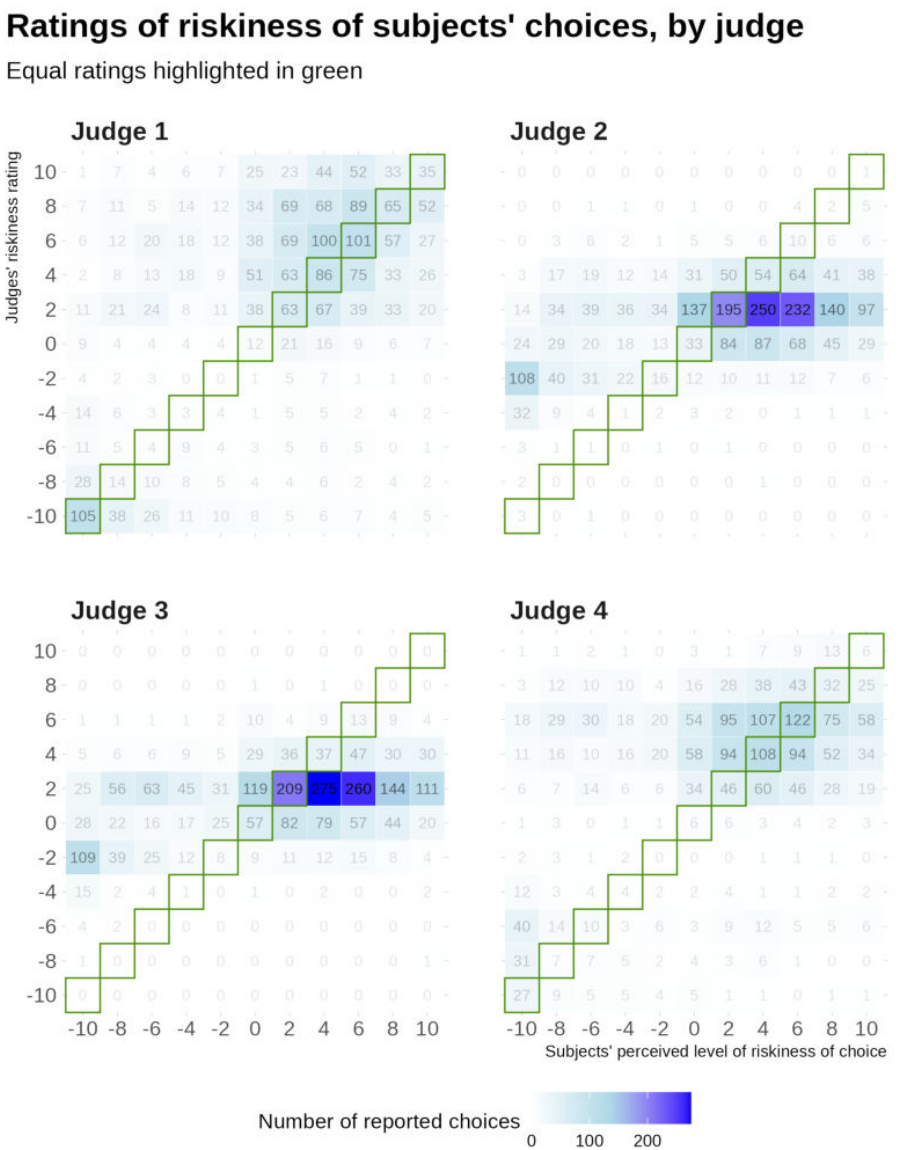
Judges' agreement rate: is the activity risky?

Judges' reliability

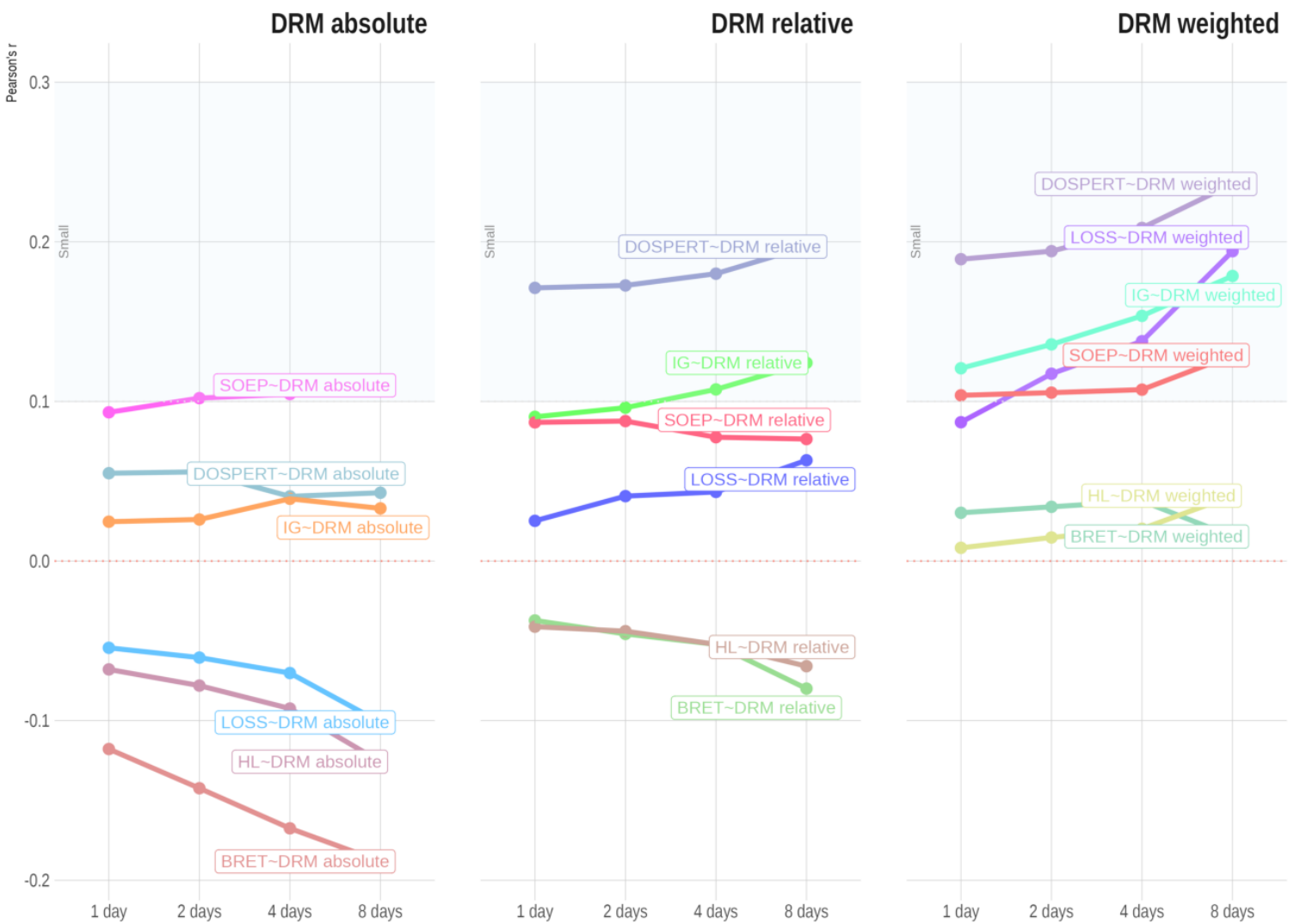
Judge1	Judge2	Mean distance	Correlation
Overall	–	3.64	0.60
Subject	Judge 1	4.31	0.55
Subject	Judge 2	4.17	0.47
Subject	Judge 3	4.22	0.44
Subject	Judge 4	4.29	0.46
Judge 1	Judge 2	4.41	0.68
Judge 1	Judge 3	4.42	0.67
Judge 1	Judge 4	3.21	0.65
Judge 2	Judge 3	1.11	0.73
Judge 2	Judge 4	3.40	0.68
Judge 3	Judge 4	3.51	0.65

Judge's agreement rate: how much risk was taken?

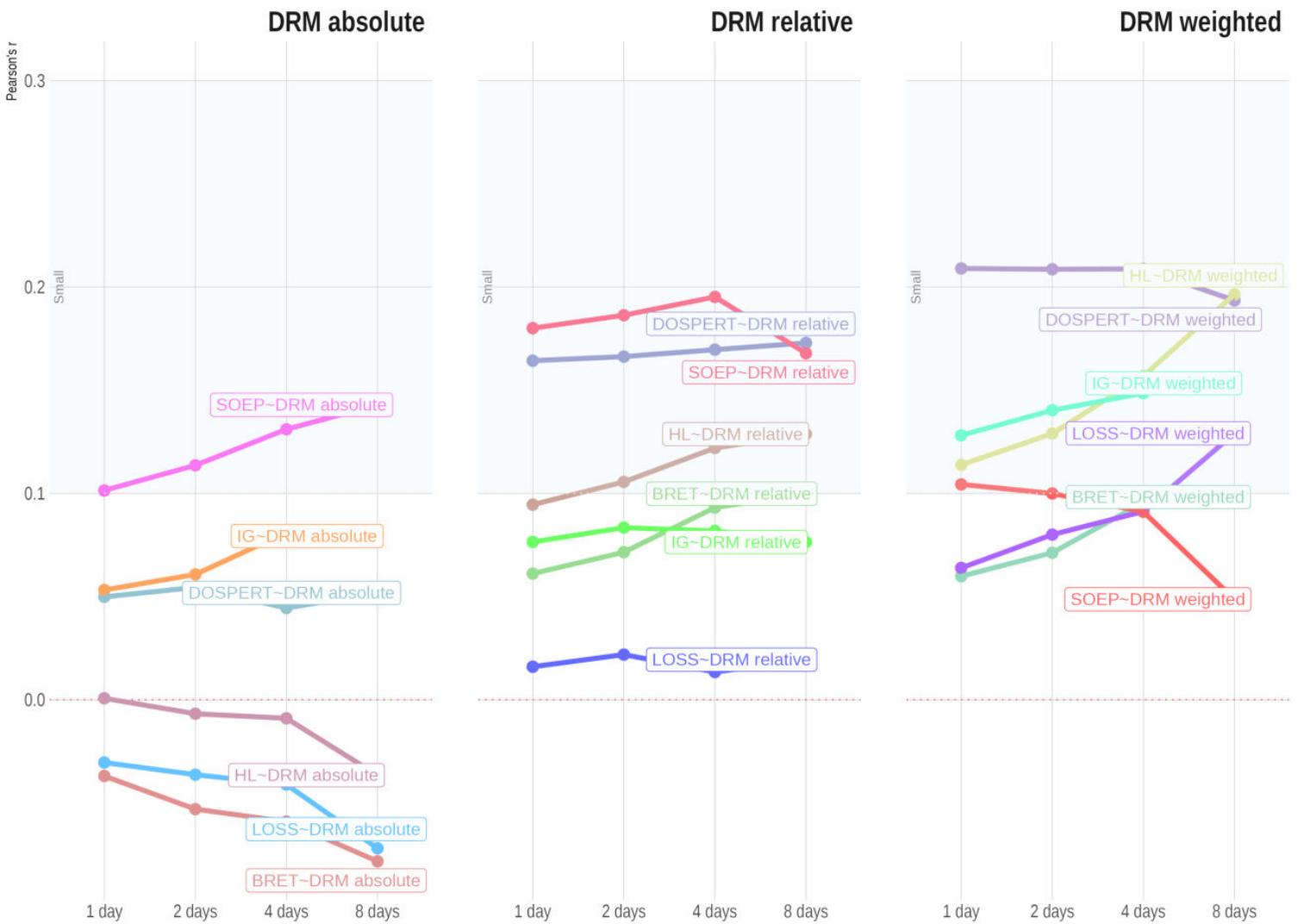
Judge’s and subject’s risk perception



External validity: aggregating (subj)



External validity: aggregating (judges)



Discussion

Not only measurement error

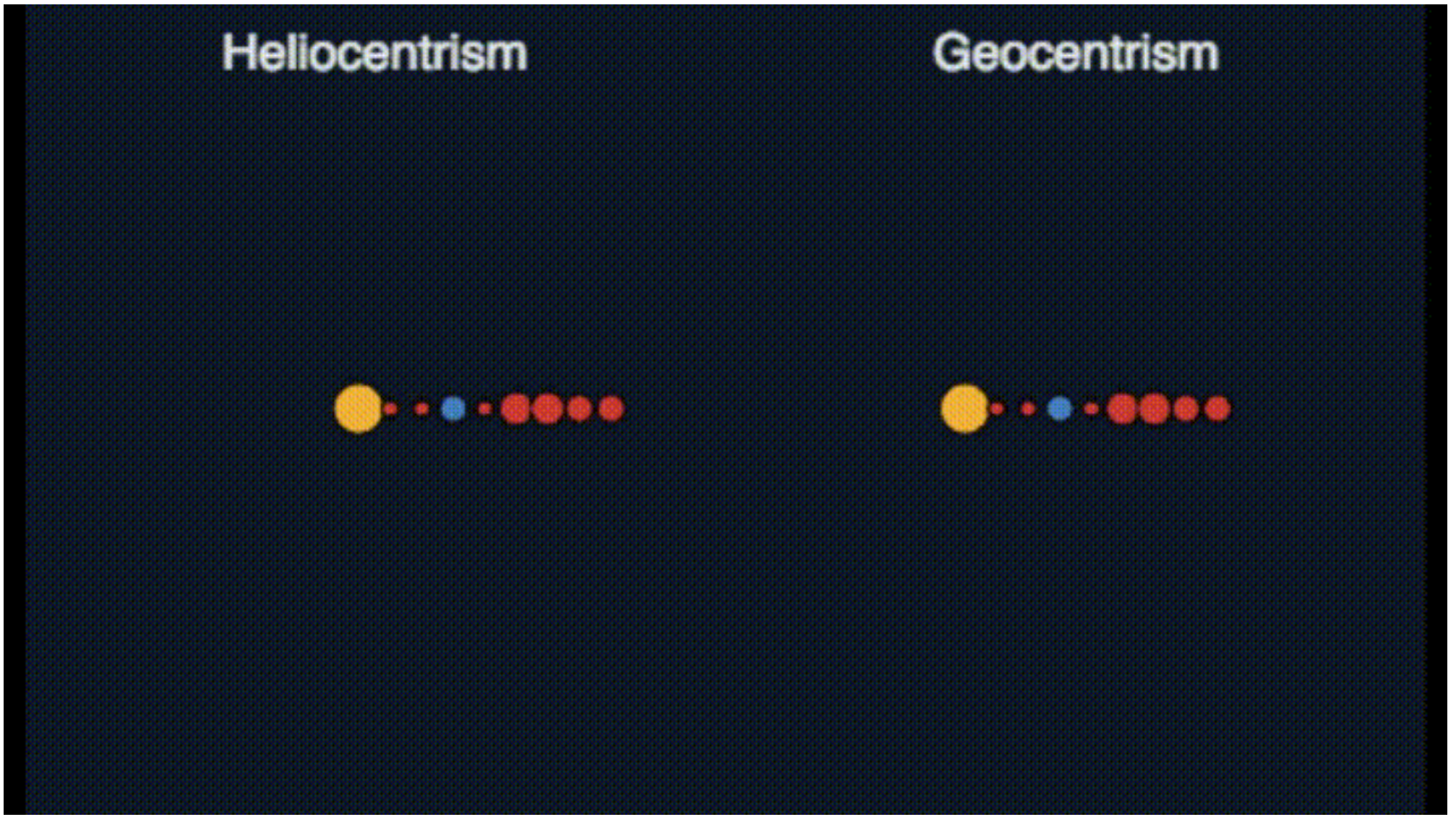
> Measurement error **plays a role**

- test-retest **dramatically up**
- convergent validity **slightly up**
- external validity **slightly up**

> But it **does not** close the risk validity gap

- **convergent** validity: tasks worse than questionnaires
- **external** validity still abysmally **low**

What if…



Have we got the right theory?

> We have so far **assumed EUT** (or **PT**) + noise. But it's no more the only game in town!

- role of **risk perception** (Holzmeister et al. ManSci 2019)
  - laymen & traders perceive risk as probability of loss
  - perception seems *not* driven by variance but skewness

Have we got the right theory?

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- role of **risk modeling** (Crosetto & Flippin 202?)
  - risk, ambiguity, deep uncertainty
  - a nested structure, a nested experiment
    - what maps better to subjects' behavior?

Have we got the right theory?

> We have so far **assumed EUT** (or **PT**) + noise. But it's no more the only game in town!

- role of **cognitive noise**
  - models of *noisy coding*: what we see as *risk aversion* could be just risk neutraility + the *way we see the world*
    - logarihtmic number perception (Khaw et al. 2021)
    - Bayesian Inference Model (Vieider 2024)
    - Role of complexity & Cognition (Oprea 2024)


TO BE
CONTINUED...

Thank you!